

NCC Distribution Management System

Diploma Software Engineering

Final Project Documentation

2025.2F

Group Members

CODSE25.2F - 006 – A.K.H Silva

CODSE25.2F - 010 – S.A.S Tharuka

CODSE25.2F - 013 – W.M.H Kumari

CODSE25.2F - 039 – T. Loshan



School of Computing and Engineering

National Institute of Business Management

Colombo - 07

Table of Contents

Chapter 1: Introduction	1
1.1 Introduction of the Organization	1
1.2 Organization Structure	2
1.3 Current Operations in Organization.....	4
1.4 Users and Responsibilities Organization	4
1.5 Definition (Problem Definition)	6
1.6 Project Objectives	7
1.7 Proposed Solution	8
Subsystems of the Proposed System:.....	8
1.8 Chapter Summary.....	14
Chapter 2: Methodology.....	14
2.1 Introduction.....	14
2.2 Methods of collecting data.....	15
2.3 Software Process Model.....	15
2.4 Software Development Tools	17
2.5 Testing Strategies	17
2.6 Implementation Plan	17
2.7 Chapter Summary.....	18

Chapter 1: Introduction

1.1 Introduction of the Organization

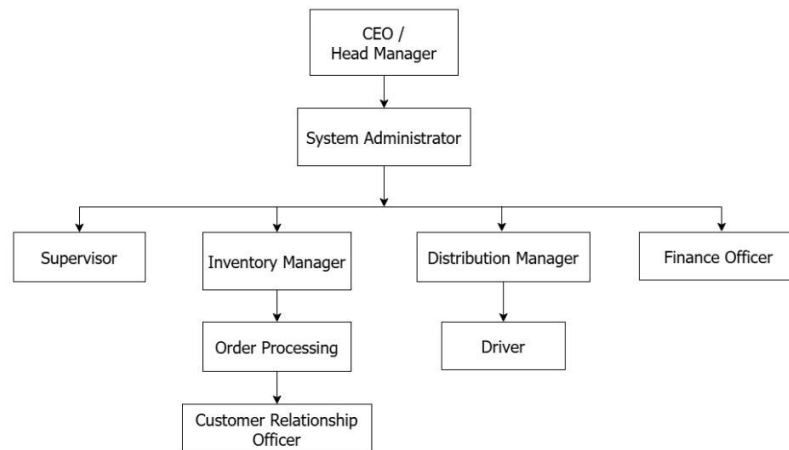
NCC is an organization operating in the agricultural supply sector, providing fertilizers and pesticides to farmers and agricultural retail vendors. The company commenced its operations in 2019 on a limited scale, focusing primarily on serving local demand.

Between 2020 and 2022, during the COVID-19 pandemic, the organization experienced financial and operational difficulties that affected its stability. Despite these challenges, NCC continued its operations and maintained its customer base through consistent service delivery. At present, the company remains active in supplying agricultural inputs to farmers and retailers.

The organization manages the distribution of its products using its own transport facilities, including three delivery trucks, which support timely product distribution. NCC intends to expand its customer reach to additional regions and is also considering the future production of fertilizers and pesticides using raw materials. These initiatives are aimed at improving supply continuity and meeting increasing agricultural demand.

Currently, NCC uses Microsoft Excel spreadsheets to record and manage inventory, sales, employee information, and financial transactions. While this method supports basic record-keeping, it has become less effective as the volume of data and daily operations has increased, leading to limitations in efficiency and data management.

1.2 Organization Structure



NCC has a clear management structure where responsibilities are divided among different roles to make daily operations easier to manage. Each person has a specific role, but all departments are connected under the supervision of the CEO.

Chief Executive Officer (CEO) / Head Manager

The CEO is the founder and owner of the organization and also acts as the Head Manager. He is responsible for the overall direction of the company. He makes major business decisions, approves budgets, and monitors the overall performance of the organization.

The CEO supervises all managers and ensures that each department is working properly. He also reviews financial performance and makes decisions related to business growth and expansion.

System Administrator

The System Administrator manages the computerized system. This person creates user accounts, assigns access levels, and makes sure the system runs smoothly. The System Administrator is also responsible for maintaining data security and performing backups when necessary.

Inventory Manager

The Inventory Manager is responsible for managing stock. This includes checking product availability, updating stock levels, and recording damaged or expired items. This role helps prevent stock shortages and ensures products are available when needed.

Order Processing Officer

The Order Processing Officer handles customer orders. This includes recording orders, checking stock availability, and generating invoices.

Customer Relationship Officer

The Customer Relationship Officer maintains customer records and handles complaints. This role ensures that customers are satisfied and that their issues are properly addressed.

Distribution Manager

The Distribution Manager plans and monitors deliveries. This includes assigning vehicles and drivers and making sure products are delivered on time.

Driver

Drivers are responsible for delivering products to customers according to assigned routes. They update the delivery status and report any issues during transportation.

Finance Officer

The Finance Officer manages financial records such as income, expenses, and payments. This role also prepares financial reports for management.

Supervisor

The Supervisor manages employee attendance and payroll preparation. This includes monitoring overtime and keeping track of daily employee activities.

Overall, the organization has a proper division of responsibilities, but it currently lacks an integrated system that connects all departments together.

1.3 Current Operations in Organization

NCC currently uses Excel spreadsheets to manage its data. Each department maintains its own separate files for inventory, deliveries, employee attendance, sales and financial records.

Inventory levels are checked manually and then updated in Excel. The Sales Manager records all sales orders by hand. Delivery routes are planned without using route optimization tools. The Supervisor manually records employee payroll and attendance is calculated separately. Because the system is not centralized, data sharing between departments is slow, causing operational inefficiencies and delayed management decision-making.

1.4 Users and Responsibilities Organization

1. CEO / Head Manager

- Oversee overall business operations
- Approve budgets and major decisions
- Monitor sales, inventory, and financial performance
- Make strategic business decisions

2. System Administrator

- Manage user accounts
- Assign system access levels
- Maintain system security
- Perform database backups

3. Inventory Manager

- Monitor stock levels
- Record stock movements
- Track shortages and damaged products
- Generate inventory reports

4. Order Processing Officer

- Process customer orders
- Generate invoices
- Verify product availability
- Update order status

5. Customer Relationship Officer

- Maintain customer records
- Handle complaints
- Communicate with customers

6. Distribution Manager

- Plan delivery schedules
- Assign vehicles and drivers
- Monitor delivery activities

7. Driver

- Deliver products to customers
- Update delivery status
- Report delivery issues

8. Finance Officer

- Record income and expenses
- Prepare financial reports
- Monitor payments

9. Supervisor

- Record employee attendance
- Monitor overtime
- Prepare payroll sheets

1.5 Definition (Problem Definition)

The Excel system we are using now is causing a lot of problems with the way things are done. The current Excel-based system has problems that are affecting our operations. For example, there is a very old Excel-based system. It has problems. Because of the Excel-based system, we have to face these problems every day.

Poor Inventory Management

The company found out about the stock shortages after it was already too late. They did not notice the stock shortages enough. This meant that the stock shortages were a problem for a time before the company did something, about the stock shortages. The stock shortages were only identified when it was too late to do anything to fix the stock shortages.

Inefficient Delivery Management

The delivery routes were not planned well. This made the whole delivery process very bad. The people, in charge of making the delivery routes did a job. They should have planned the delivery routes better. The delivery routes were poorly planned.

The deliveries were delayed. This happened with some of the deliveries. The deliveries did not arrive on time.

The company wasted a lot of fuel and time because the routes that the drivers took were not very good. The routes were not planned in a way that would save fuel and time. This meant that the drivers had to drive a lot more than they needed to which wasted a lot of fuel and time. The fuel and time that were wasted could have been used for something if the routes had been better planned.

Payroll Issues

The company did not always get overtime payments right. Sometimes they made mistakes when they figured out how money people should get for working extra hours.

Overtime payments were wrong. That was a problem.

Attendance tracking was inconsistent.

Customer Service Problems

The company did not handle customer complaints in a way. Customer complaints were a problem and the company did not do a good job of solving them. The customer complaints were not taken care of quickly. That was a mistake. Customer complaints are very important for the company so the company should have done a better job of handling customer complaints.

Order tracking was not systematic.

Data Management Issues

The data duplication problem happened a lot. We had to face with data duplication all the time. This issue, with data duplication was really frustrating because it kept happening over again with the data duplication.

High risk of accidental deletion or modification.

No proper backup system.

These problems highlight the need for a computerized system.

1.6 Project Objectives

The main goal of this project is to create a NCC Distribution Management System. This new system will replace the system that uses Excel. The new NCC Distribution Management System will be Developed from scratch.

Specific objectives:

- Inventory tracking
- Improve order complaint handling and processing
- Optimize delivery route planning

- payroll calculations and Automate attendance
- Centralize all data into a secure MySQL database
- Generate real-time analytical reports
- Support future business expansion

1.7 Proposed Solution

We propose to develop a **Web-Based Distribution Management System** using:

- **Front-End:** HTML, CSS
- **Back-End:** PHP
- **Database:** MySQL
- **Deployment:** XAMPP
- **UI/UX Design:** Figma
- **Version Control:** GitHub
- **SDLC Model:** Waterfall Model

Subsystems of the Proposed System:

1. System Administrator

Functional Requirements :

- Secure login and logout
- Create new user accounts
- Assign user roles
- Define access privileges
- Update user information
- Reset passwords
- Activate or deactivate accounts
- Monitor user activity logs
- Track login history
- Configure system settings
- Manage notification settings
- Perform database backups

- Restore system data
- Monitor system errors
- Generate audit reports

2. CEO / Head Manager

Functional Requirements :

- Secure login and profile management
- Access centralized business dashboard
- View real-time sales performance
- View inventory valuation
- View stock movement reports
- View delivery performance reports
- View profit and loss statements
- Monitor revenue growth trends
- Approve budgets and financial plans
- Approve major purchases
- Approve discount policies
- Assign departmental targets
- View employee performance reports
- Generate strategic reports
- Monitor system-wide activities
- View audit logs
- Handle escalated complaints
- Send internal announcements
- Approve business expansion plan

3. Inventory Manager

Functional Requirements :

- Add, categorize, update, and delete products
- Set product pricing and minimum stock levels
- Record incoming stock and batch numbers
- Track expiry dates and damaged/expired items

- Record stock returns and transfer between locations
- Generate stock valuation and movement reports
- Receive automatic low-stock alerts
- Monitor inventory turnover

4. Order Processing Officer

Functional Requirements :

- Create, edit, and cancel sales orders
- Verify stock availability
- Approve credit orders
- Generate invoices
- Apply discounts and calculate taxes automatically
- Update order status
- Generate order reports
- Track daily sales totals

5. Customer Relationship Officer

Functional Requirements :

- Register and update customer details
- View customer purchase history
- Maintain client database
- Record and respond to complaints
- Track complaint resolution status
- Manage loyalty programs

- Send promotional notifications
- Generate customer activity reports

6. Distribution Manager

Functional Requirements :

- View confirmed orders
- Plan delivery schedules
- Assign drivers and vehicles
- Optimize delivery routes
- Track dispatched orders
- Monitor delivery progress
- Record transportation costs
- Generate delivery performance reports
- Identify delayed deliveries

7. Driver

Functional Requirements :

- Secure login
- View assigned delivery routes and delivery details
- Update delivery status and confirm completed deliveries
- Report delivery issues
- Upload proof of delivery
- Record fuel usage

8. Supervisor

Functional Requirements :

- Record employee attendance
- Monitor clock-in and clock-out times
- Manage overtime records
- Approve leave requests
- View employee details
- Calculate salaries automatically
- Prepare payroll sheets
- Generate salary slips
- Track employee performance
- Generate attendance reports

9. Finance Officer

Functional Requirements :

- Record daily income and operational expenses
- Manage supplier payments
- Track outstanding receivables
- Generate profit and loss statements
- Generate cash flow reports
- Prepare financial summaries
- Monitor budget utilization
- Maintain tax records
- Perform account reconciliation

Non-Functional Requirements

1. Product Requirements

- The system should provide secure authentication with encrypted passwords.
- The system should maintain audit logs for accountability.
- The dashboard shall load within 3 seconds under normal operating conditions.
- The system should provide real-time data updates across all modules.
- The system shall support multiple concurrent users without performance degradation.
- Backup and restore operations shall not interrupt normal system functionality.
- The system shall ensure high availability with minimum 99% uptime.
- Financial, tax, and payroll calculations shall be accurate.
- Delivery tracking updates shall be near real-time.
- The database shall maintain data consistency and integrity.
- The mobile interface should be responsive and user-friendly.
- The system shall allow reports to be exported in PDF and Excel formats.

2. Organizational Requirements

- The system shall follow NCC's internal access control policies.
- User roles and privileges shall be assigned according to organizational hierarchy.
- Training shall be provided to employees prior to system deployment.
- The system shall support future business expansion.
- All organizational data shall be centralized within a single secure database.

3. External Requirements

- Payroll calculations shall comply with Sri Lankan labor laws.
- Financial reporting shall comply with applicable accounting standards.
- Customer data shall be handled in accordance with data protection regulations.
- Tax calculations shall comply with government tax regulations.

4. Domain Requirements

- The system shall track product batch numbers and expiry dates.
- The system shall generate low-stock alerts to prevent shortages.
- The system shall support inventory valuation suitable for agricultural products.
- The system shall support route optimization for distribution efficiency.

1.8 Chapter Summary

An introduction to NCC Fertilizer & Agro Supplies and its overall structure and current operations were described in this chapter. Current operational problems related to the existing Excel-based system were identified. Objectives of the new Distribution Management System were identified and it is proposed that a web-based solution be developed as part of this chapter. This chapter also sets the stage for the development of NCC's Modern Distribution Management System.

Chapter 2: Methodology

2.1 Introduction

This chapter includes each method that will be used to create the NCC Distribution Management System, including how the Waterfall Software Development Life Cycle will be implemented to build the system.

Development process contains the following steps:

- 1) Gathering Requirements**
- 2) Designing the System**
- 3) Development**
- 4) Testing**
- 5) Deployment**
- 6) Maintenance**

2.2 Methods of collecting data

1. Interviews

We conducted an interview with the Managing Director and the questions asked were:

- "What are the roles of the employees in the company?"
- "What method do you use to record your financials?"
- "How are the reservations/orders made?"
- "Where is the data stored?"
- "What kind of system would you like to have?"

The Managing Director did not allow us to view the records, however, through the interview alone, we are given a good perspective about the business processes.

2. Interviewing the Staff

We conducted an interview with the company's employees as well and the following questions were asked:

- "How do you request a leave?"
- "How are the tasks assigned to you?"
- "Are you experiencing any problems with the existing system?"

2.3 Software Process Model

The Waterfall Model has been chosen because:

- There is a clear definition of the requirements
- Stable project scope
- Structured documentation
- Each phase must be completed before moving on to the next stage

This model needs proper planning along with stable structure and NCC provided us the needed documentation



2.4 Software Development Tools

Development Technologies:

- **HTML & CSS** – Front End interface
- **PHP** – Server-Side scripting
- **MySQL** – Database management
- **XAMPP** – Local deployment server

Design Tools

- **Figma** – UI/UX design and prototyping

Version Control

- **GitHub** – Team collaboration and code management

2.5 Testing Strategies

The following strategies for testing the system are proposed:

Unit Testing:

- Testing the individual modules (inventory, sales, payroll).

Integration Testing:

- Testing the modules in an integrated manner.

System Testing:

- Testing the entire system functionality.

User Acceptance Testing:

- Allowing the system to be tested by the managers and employees.

2.6 Implementation Plan

The following are the steps to be followed while implementing the system:

1. Requirement analysis and documentation
2. Database design with ER diagrams
3. System interface design with Figma
4. Coding of front-end and back-end modules
5. Unit testing of the modules
6. Integration testing
7. Deployment with the help of XAMPP
8. User training
9. System maintenance and update

Training will be provided to the managers and the employees before the actual implementation.

2.7 Chapter Summary

This chapter has provided the reader with the methodology of how the NCC Distribution Management System will be developed. It covered the process of collecting the required data, the Waterfall Software Development Life Cycle, the development tools, the testing process and the implementation plan. It ensures the development of the system in the most efficient manner, with thorough testing and implementation, to support the business growth of NCC.